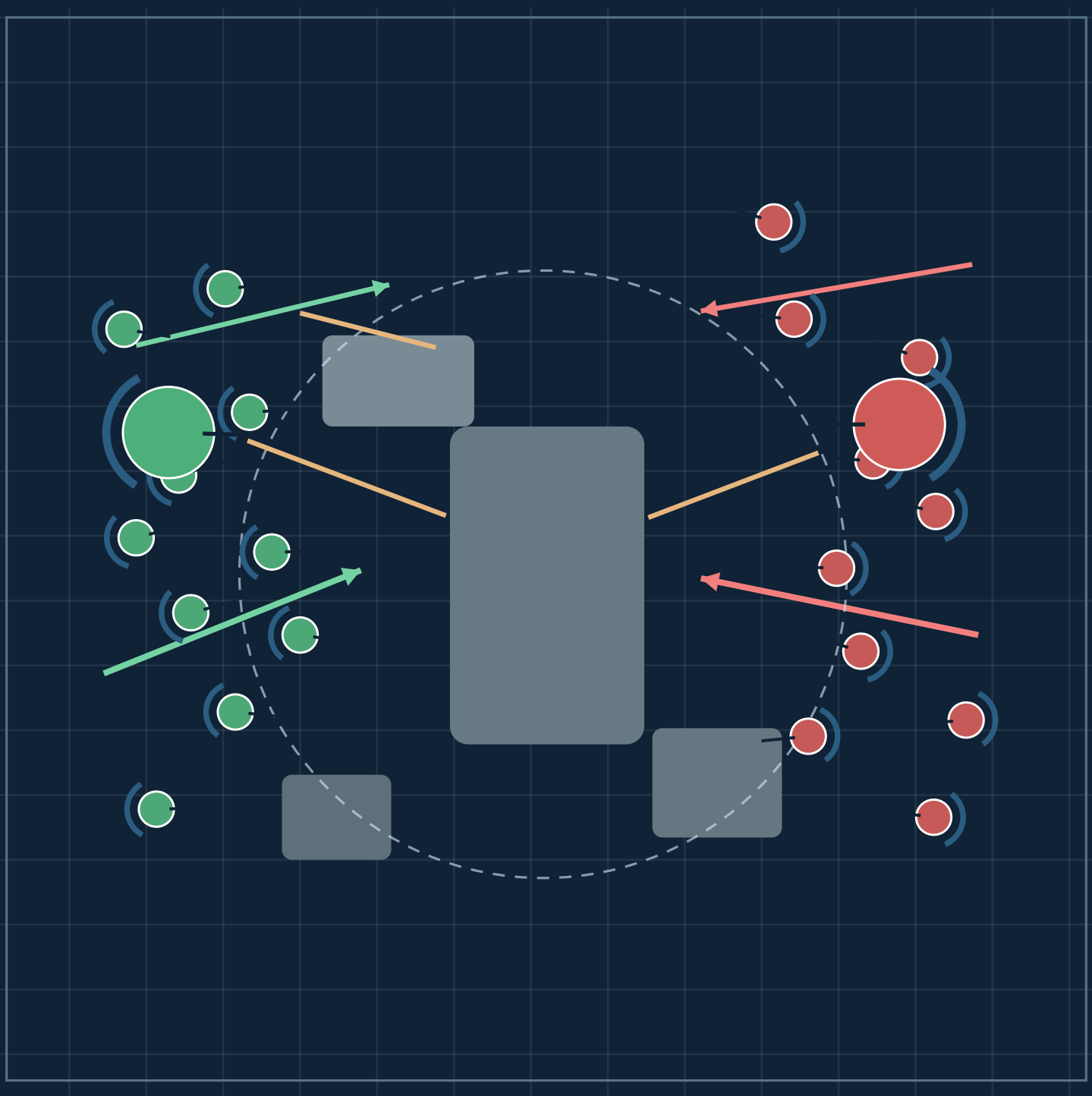
GangUP

GAME DESIGN DOCUMENT

Version 0.3

Hybrid Tactical Action and Intelligent Squad Warfare

AI Integration: SLAI - Self-Learning Artificial Intelligence



OCTOBER 2026 | PRE-PRODUCTION SPECIFICATION

Original vector cover illustration: tactical confrontation, formations, shields and cover

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A self-contained Official GangUP Ruleset follows Chapter 19. All balance figures are provisional unless explicitly identified as authoritative alpha constants.

1. Introduction and Game Overview

1.1 Identity and design mandate

GangUP is a two-faction, top-down, real-time hybrid of direct Main-character combat and explicit soldier squad command. Each force has one Main, four initial soldiers, health-dependent reinforcement, and a maximum of 50 living soldiers. The Main carries a combined rifle/sword and wears a directional rear shield. Win by defeating the opposing Main, obtaining a valid surrender, or causing Last Stand Collapse.

The design requires comparable importance from individual action skill and squad strategy. This does **not** require each system to inflict 50% of damage. Instead, controlled tests must establish that improving either player skill dimension materially improves match outcomes and that strong combined play exceeds strong one-dimensional play.

Document control	Value
Specification	GangUP Game Design Document
Version / status	0.3 / implementation-oriented pre-production baseline
Mode / perspective	Human versus SLAI AI / top-down 2D
Source priority	Original GangUP.pdf; v0.2 GDD; explicit v0.3 mandates
Document typography	Tinos embedded as an explicitly disclosed Times New Roman metric-compatible fallback; Times New Roman was not available.
Numbers	Alpha constants proposed for reproducible testing; not validated by playtest

1.2 Source continuity and scope control

Retained from the original three-page concept: red and green factions, one Main per faction, up to 50 soldiers, front rifle/sword, rear shield, health-based spawning, Q/E stances, Defend/Follow/Press/Harass/Disengage, and defeat/surrender/low-health collapse. The v0.2 design clarified health, ammunition, cooldowns, squad slots, and separate movement and combat commands. v0.3 adds validated line of sight, accuracy, streak modifiers, and strict backward-speed rules.

No repository is changed by this document. The Game Engine, local soldier controller, squad tactical controller, and SLAI are **design interfaces**; they are not represented as already implemented.

1.3 Terminology and notation

Term	Meaning
Main	Unique commanding character of a faction.
Soldier (Minion)	Autonomous combat unit serving one Main; the terms are synonyms.
Squad	Player-managed grouping of no more than ten soldiers.
Doctrine	A persistent combat behavior: Defend, Press, Harass, or Disengage.
Order	Movement/follow/hold, targeting, formation, or combat doctrine instruction.
LoS	Line of sight: geometric visibility independent of projectile collision.
Reinforcement Wave	Scheduled 20-second soldier arrival opportunity.
Last Stand Collapse	Loss after 30 continuous seconds with Main HP below 10% and zero living soldiers.

Specification convention: MUST and MUST NOT are invariants; SHOULD and MAY mark guidance or options. All timings use simulation time.

2. Core Gameplay Philosophy

2.1 Player decision loop

The player repeatedly observes visible threats and squad status, chooses a movement orientation and weapon, acts in real time, and delegates persistent tasks to squads. Combat success creates tactical opportunities; squad positioning creates safe approaches and favorable engagements for the Main.

The enemy follows equivalent rules but makes its decisions through a hierarchy of SLAI strategic planning, squad tactical control, and fast local soldier execution. The player and SLAI receive symmetric information through the same visibility rules.

2.2 Five design pillars

Action/strategy parity: both direct fighting and squad orders matter. **Distinctive orientation:** front weapon and rear shield produce nontraditional protection decisions. **Autonomous execution:** soldiers carry out high-level orders, not per-frame instructions. **Fair intelligence:** SLAI has no stat bonuses or hidden omniscience. **Readable pressure:** the interface explains every change in soldier wave size, shield bonus, and tactical state.

2.3 Tactical trade-offs

Rifle attacks spend finite ammunition and require a clear lane; sword attacks have no ammo expenditure but raise strain and expose the attacker. A healthy Main enjoys more reinforcements but becomes a valuable target. Defensive stance uses up to five unassigned bodyguards while offensive stance reserves two. Dedicated squads remain independently commanded in either stance.

2.4 Measurable hybrid fairness

Evaluate matched seeded pairs using action-only, strategy-only, combined, and baseline policies. Report victory rate with confidence intervals and compare identical maps with factions swapped. Combined policies should outperform single-specialty policies, but rare individual upsets remain desirable. Only controlled match data can substantiate the objective.

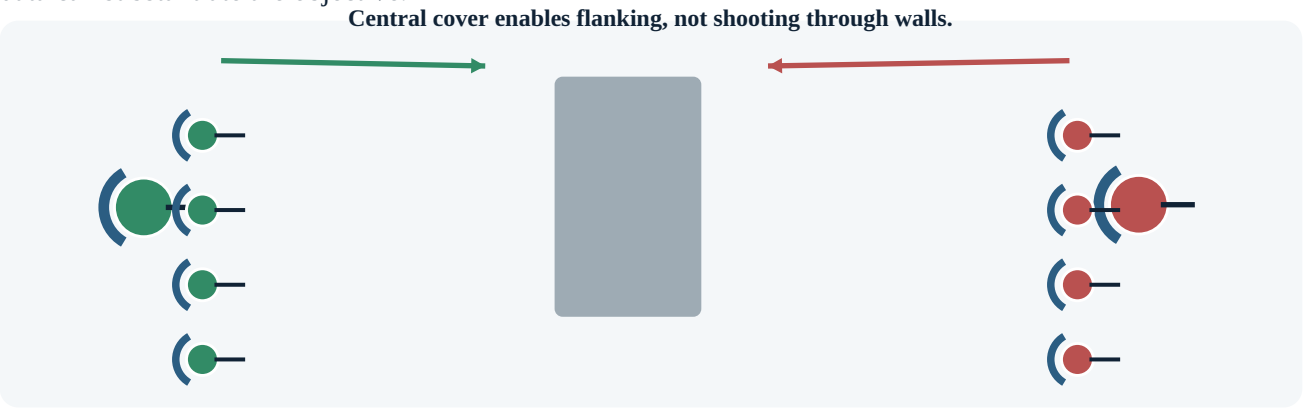


Figure 1. *Tactical space: two factions approach a centrally obstructed region with lateral options.*

3. Game Entities and Character Attributes

3.1 Faction and entity definitions

Exactly two opposed teams exist in the initial mode. Team Green is the human side; Team Red is controlled by SLAI. Visual labels supplement color for accessibility. "Defeated" is a nonactive state, not a third team. Neutral units/objects are nonhostile and nonrecruitable in v0.3; unrelated recruitment mechanics are excluded.

Attribute	Main	Soldier
Maximum body HP	100	30
Base shield durability	25	8
Forward speed (m/s)	4.0	6.0
Backward speed (m/s)	2.8	4.2
Rifle and sword damage multiplier	1.00	0.20
Attack-based healing	Eligible	None
Command authority	Allied squads	Execute validated orders
Collision body radius (provisional)	0.50 m	0.35 m

3.2 Representation and precision

HP, shield HP, strain, and displacement use consistent numerical precision internally. UI rounding MUST NOT change authoritative thresholds. Body HP is bounded [0, maximum]; shield HP is bounded [0, current shield maximum]; strain is bounded [0,100]. Depleted shields cease to intercept damage until durability regenerates above zero. A defeated character cannot deal later attacks or issue new commands.

3.3 Entity ownership and stable IDs

Each active unit has a match-unique identifier and permanent faction ownership. Group IDs are distinct from unit IDs. Soldier removal invalidates squad-leader references and any active target reference pointing to the defeated unit. The game must not recycle IDs within a match because queued AI orders and replay events can reference earlier units.

3.4 Spawn/death invariants

Four soldiers begin per team, and each team can have no more than 50 living soldiers. A newly spawned soldier starts with 30 HP, 8 shield HP, a 6-round rifle magazine and 18 reserve rounds (24 carried total), zero sword strain, and an available initial command state. Soldier death is permanent for the current soldier identity; subsequent reinforcement creates a distinct soldier.

4. Movement, Rotation, and Shield Orientation

4.1 Tank controls and independent vectors

W moves forward in the facing direction; S moves backward along the opposite direction; A/D rotate counterclockwise/clockwise. The front weapon and rear shield rotate **only** with the character. Movement direction can be opposite the facing direction. An actor cannot independently swivel its shield or rifle in v0.3.

Facing is a normalized angle theta. Weapon forward is (cos theta, sin theta). Rear shield center is theta + 180 degrees. Forward movement uses +facing vector; backward movement uses -facing vector. Simultaneous rotation and translation are allowed.

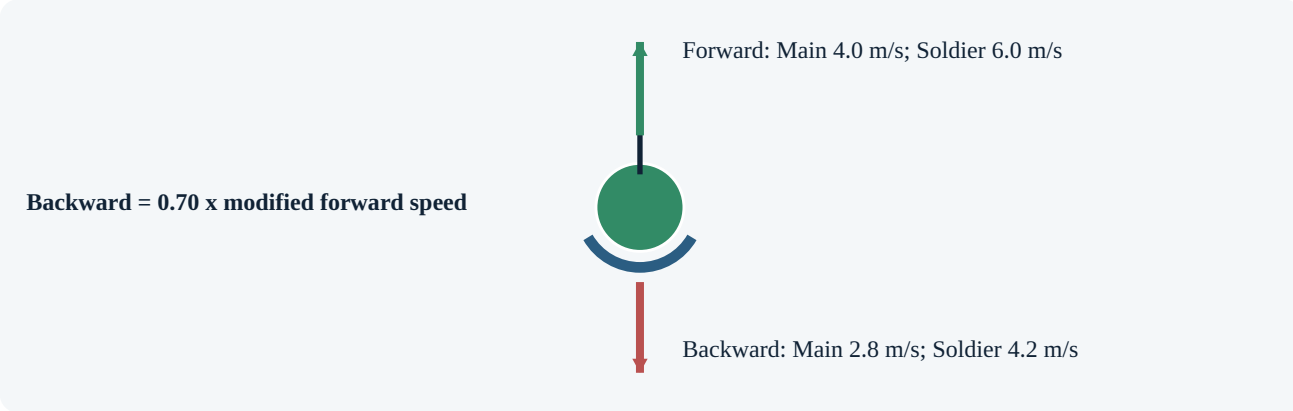


Figure 2. *The mandatory 30% backward-speed reduction applies to Main and soldier.*

4.2 Exact speed calculation order

Define base forward speed V (4 or 6 m/s) and the currently active **doctrine speed multiplier** M (default 1.00; Press 0.85; Harass 1.25). Calculate modified forward speed F = V x M. Calculate backward speed B = F x 0.70. Finally apply movement-direction sign to velocity. No double application of the 30% penalty is permitted.

Entity / doctrine	Forward m/s	Backward m/s
Main / neutral doctrine	4.00	2.80
Soldier / normal	6.00	4.20
Soldier / Press	5.10	3.57
Soldier / Harass	7.50	5.25

4.3 Acceleration, rotation, and collision

Provisional acceleration limits are 12 m/s² for the Main and 16 m/s² for a soldier; braking uses the same magnitude. Initial Main angular speed is 180 degrees/s; soldier maximum is 240 degrees/s. The speed modifier changes translational speed but not angular speed. Engine collision resolution applies movement against solid static obstacles and neighboring units; no actor tunnels through walls or accumulates unbounded displacement.

Each unit separates from nearby allies using deterministic collision response, ordered by stable actor ID when ties occur. Congestion can slow units but never raises their top speed. Orders remain pending when a path is temporarily blocked; repeated failure triggers a bounded replanning and fallback response, not infinite spinning.

5. Weapons, Rifle Accuracy, and Combat Resolution

5.1 Rifle baseline and exact damage function

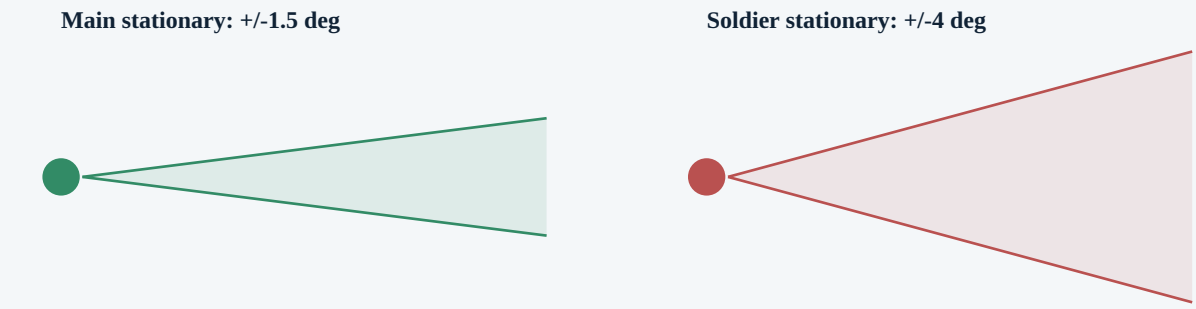
The Main rifle fires one round every 0.8 seconds, with 10 rounds in its magazine, up to 50 rounds carried including the magazine, and a 1.5-second reload. Soldier rifle fires every 1.5 seconds, magazine 6, maximum carried 24, reload 1.5 seconds. Main bullet damage is 15 HP for hit distance d <= 2 m; for 2 < d <= 18 m it is **15 - 13(d - 2)/16**; beyond 18 m the shot misses by range. Soldier body/shield damage is 0.20 times the unmodified Main bullet damage before doctrine and shield modifiers.

Range	Main bullet damage	Soldier bullet damage
0-2 m	15.000	3.000
6 m	11.750	2.350
10 m	8.500	1.700
14 m	5.250	1.050
18 m	2.000	0.400
>18 m	0 (out of range)	0 (out of range)

5.2 Dispersion semantics

Rifle spread values are **maximum signed angular deviations**, not standard deviations or probabilities of hitting. For each fired round, theta-shot = theta-aim + u(-a,+a), where u is drawn uniformly from a deterministic match-seeded pseudo-random stream and a is the shooter spread half-angle. No AI receives a separate accuracy exception. A shooter is "moving" when actual translational speed exceeds 0.2 m/s at the fire tick; rotation alone does not select moving spread.

Shooter	Stationary +/-	Moving +/-
Main	1.5 degrees	2.5 degrees
Soldier	4.0 degrees	6.0 degrees



Visualization exaggerates angular differences; spread is seeded and uniform.

Figure 3. *Uniform angular half-angle dispersion; visual wedges are schematic, not to scale.*

5.3 Shot creation and first-hit resolution

Rifles use authoritative instantaneous raycasts (hitscan) in v0.3. On a legal fire tick the engine verifies weapon mode, alive state, cooldown, ammo and no conflicting reload; consumes one magazine round; samples seeded spread; casts from the forward muzzle along the resulting direction to 18 m; then resolves the **nearest** collision with opaque bullet-blocking geometry, an allied body, an enemy shield or an enemy body. Obstacles absorb shots. Allies absorb shots harmlessly because friendly damage is off. A clear shot may still miss because of spread or target motion before the firing tick.

SLAI soldier controllers SHOULD refrain from firing when predicted muzzle-to-target geometry intersects an ally first. Human manual fire is permitted, but an allied collision wastes the round with no friendly damage. Hitscan does not grant projectile travel-time leading; target positions are evaluated from the authoritative firing tick. A projectile cannot penetrate another actor or wall.

5.4 Reload, sword mode and interruptions

Reload begins only when a magazine is not full and reserve ammunition exists. Magazine rounds are moved from reserve only at the completion of the 1.5-second uninterrupted reload. A weapon switch, sword attack, defeat, or match end cancels the reload without consuming reserve rounds. Movement and rotation do not cancel reload. Shots attempted during reload fail without consuming ammunition. Fire cooldown persists across weapon changes and cannot be reset by switching.

5.5 Sword attacks and shared rules

Sword property	Wide Swipe	Focused Thrust
Main damage	7 HP	12 HP
Soldier damage	1.4 HP	2.4 HP
Attack reach	2.0 m	2.5 m
Cooldown	0.85 s	0.85 s
Max victims	3	1
Strain on successful hit	18 points	24 points

Swipe evaluates the closest three valid intersecting enemies within a frontal 120-degree sector. Thrust evaluates the nearest valid intersecting enemy within a narrow 20-degree forward sector. Ties use stable actor IDs. A target completely separated by solid geometry is ineligible even if within geometric range. Every sword hit can be intercepted by the target's directional shield based on the attack direction. The shared sword cooldown applies to both sword variants, cannot be reset by switching, and prevents duplicate hits from one swing.

5.6 Sword strain lifecycle

An eligible hit adds strain once per attack attempt, not once per target. A complete miss adds zero strain; a shield interception still counts as a successful hit. Recovery starts one second after the latest sword attack attempt and subtracts 12 points/s. When strain reaches 100, sword attacks cease for 6.0 simulation seconds, then strain is set to 40. Rifle use remains possible during overload. Sword attacks already resolved on the overload-triggering tick remain valid.



Figure 4. *Order of authoritative geometry, damage, healing, streak, reinforcement and match checks.*

6. Line-of-Sight, Visibility, and Environmental Obstacles

6.1 Deterministic perception geometry

The initial arena uses solid, opaque, bullet-blocking rectangles/polygons; no transparent bulletproof cover exists in v0.3. Each Main and soldier has a 24 m vision range and a 220-degree forward-centered field of view (FOV). The FOV is based on actor facing. An enemy outside FOV or beyond sight radius is not currently visible, even if the area is visible to the game camera.

For a circular target, LoS samples its center and eight evenly spaced boundary points placed just inside the collider. The target is considered **partially visible** if at least one point is in range, inside FOV, and connected to the observer center by an unobstructed ray. A fully occluded or otherwise unsighted target is **not visible**. Samples are a deterministic and conservative approximation, not literal pixel visibility.

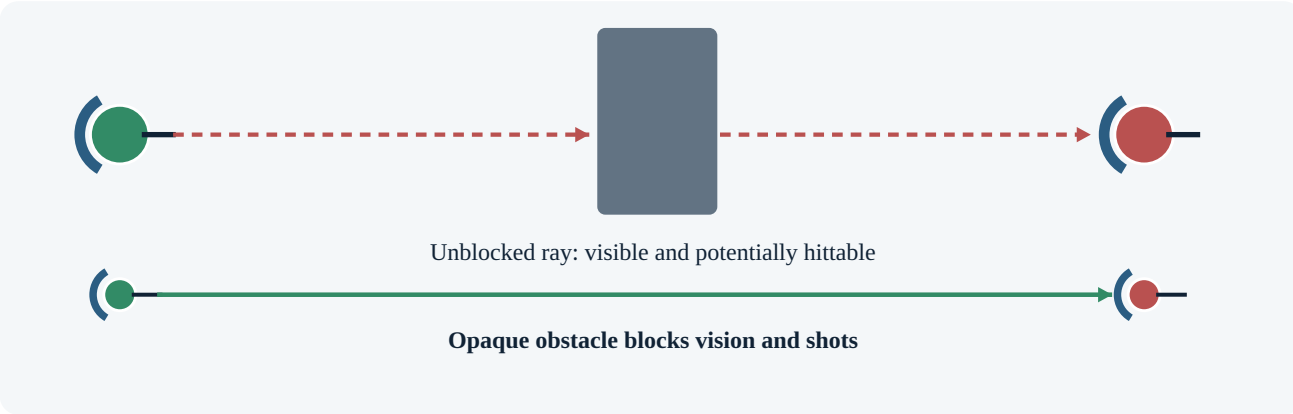


Figure 5. *Opaque cover blocks both sight rays and shots; clear rays can reveal and engage a target.*

6.2 Visibility versus hit eligibility

Visibility is an information rule; physical damage is an independent geometry rule. The actor must have current line of sight to automatically acquire and track an enemy. A rifle shot must also have an unobstructed muzzle ray after spread; a visible enemy can still be missed, blocked by an ally, or protected by a shield. For sword hits, the attack sector and collision path must be clear. A blind human shot may be fired toward a location, but cannot penetrate solid cover and does not grant information about a concealed target.

6.3 Last-known-position memory

When a previously visible enemy becomes concealed, its exact position in the observer's tactical observation is replaced by a timestamped last-known position that remains eligible for investigation for **3.0 seconds**. It is frozen, does not update through walls, carries "stale" status, and is never a direct auto-attack target. After the three seconds expire, the observation records the opponent as unknown until reacquired. At expiry, stale positional UI indicators disappear.

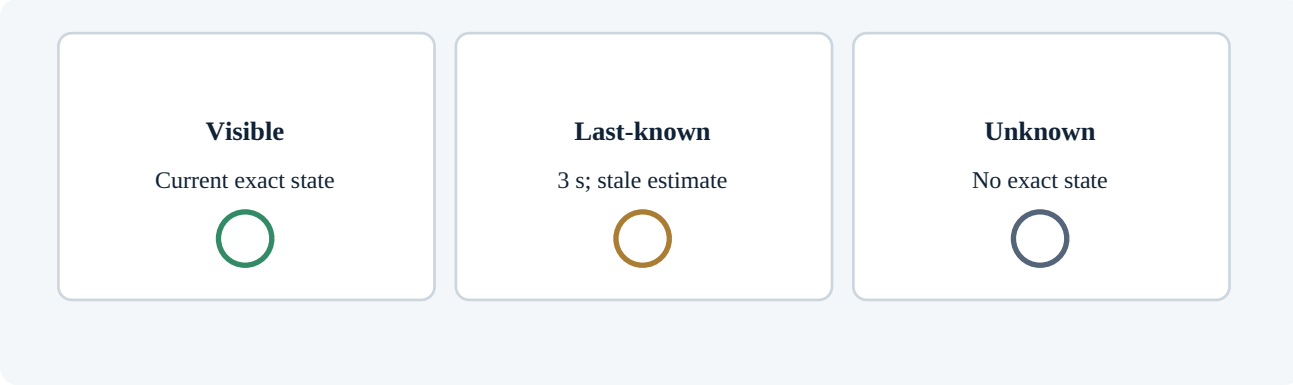


Figure 6. *Visibility states distinguish current exact sightings, stale memory and unknown locations.*

6.4 Team sharing and UI parity

A team's tactical shared view consists of the union of **current valid sightings** from its living Main and soldiers. Sightings may be shared through the same deterministic observation stream for player and AI. Shared last-known sightings expire three seconds after the most recent legitimate observation. The top-down player camera may display terrain but MUST conceal actual hidden enemy sprites and exact HP/position. A stale marker may display the frozen last-known location with its age.

6.5 Geometry and edge cases

When sight and bullet intersections occur at the same distance, solid cover wins ties within a geometric epsilon of 1 mm. Observers cannot see or shoot through diagonal corner contacts by exploiting floating-point drift. Moving walls are out of initial scope. Partial cover does not automatically reduce damage: the actual shot ray either intersects a valid exposed target region or hits cover. Collision, shield and obstacle components use a common spatial query layer to minimize disagreement between LoS and firing.

Validation scenario	Expected outcome
Fully hidden enemy behind wall	Not visible or auto-targetable; shot blocked
Boundary point exposed	Partially visible; still requires physical hit ray
Observer and target move	Recalculate sight on permitted perception tick
AI target goes behind cover	Frozen last-known position <=3s; no perfect tracking
Two overlapping walls	Nearest opaque intersection blocks
Ally crosses firing lane	AI holds or repositions; human shot absorbed harmlessly
Enemy reacquired	Current sighting replaces stale position immediately

7. Health, Shields, Healing, and Collision Mechanics

7.1 Rear shield angle and interception

The shield protects a 120-degree rear-facing arc, i.e. a half-angle of 60 degrees around facing + 180 degrees. Let f be the defender forward unit vector and i the unit vector from defender center toward the incoming attack origin. The hit is rear-protected when **dot(f, i) <= -0.5** and shield HP > 0. For contact geometry, use collision normal analogously. The shield never independently rotates. Both Main and soldier use the same rule.

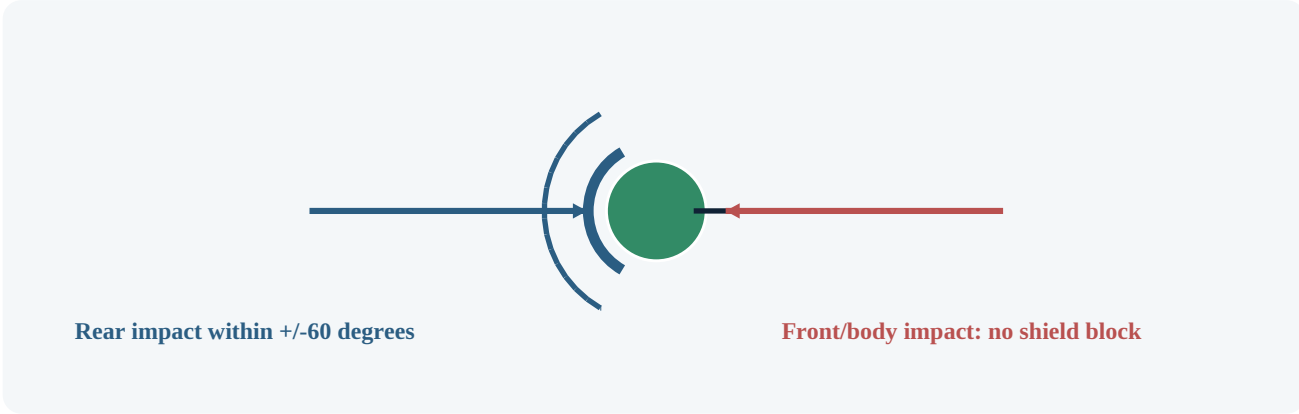


Figure 7. A 120-degree rear arc blocks incoming attacks from behind but leaves the forward sector exposed.

7.2 Shield arithmetic and recharge

Main shield maximum is 25 HP; soldier shield maximum is 8 HP. A blocked attack normally reduces shield durability by the intercepted raw damage. If current shield HP is insufficient, remaining unabsorbed damage passes to body HP. Shield regeneration begins after 8.0 continuous seconds with no attack attempt, incoming damaging contact, or successful damage received. Main regenerates 2.5 shield HP/s; soldier regenerates 1.0 HP/s. Attacking resets the regeneration delay even on miss; defensive movement does not.

Press increases **incoming shield durability consumption** by 25%, not raw incoming body damage. For a blocked raw hit D and shield consumption multiplier k, the attempted shield loss is kD. The shield absorbs fraction min(1, shieldHP/(kD)) of D; unabsorbed fraction of D damages the body. This avoids creating body damage larger than the original attack solely through a shield durability modifier. Shield never has negative durability.

7.3 Temporary shield max changes

When the 40-second streak reward activates, the Main shield maximum increases from 25 to 30 and current shield increases by up to 5 (clamped at 30). Normal regeneration applies up to the temporary max. On expiry, maximum returns to 25 and current HP is clamped to 25; the removed maximum does not turn into body damage. Reapplying or extending the shield buff is forbidden during active effect and cooldown.

7.4 Collision events

Enemy contact	Single eligible contact result
Unshielded body vs body	Both bodies lose 5 HP
Valid rear shield vs shield	Each participating shield loses 1 shield HP
Valid rear shield vs unshielded body	Shield loses 1 shield HP; unshielded body loses 5 HP
Body vs solid wall	Movement blocked; no contact HP damage
Friendly actor collision	Separation only; no collision damage

Impact damage occurs on an initial distinct collision event and only again after at least 1 second **and** a separation/re-contact threshold of 0.1 m. Repeated frame overlap never causes repeat damage. A shield at zero HP cannot protect a shield-to-body contact. A pair crossing the 120-degree boundary is classified from contact

orientation at the impact tick. Shield and body interactions are reciprocal where applicable, but one contact produces exactly one event.

7.5 Damage-based Main healing

After simultaneous incoming damage and death determination, a **surviving Main** may heal from enemy body HP actually lost to its own rifle or sword on that tick. Shield damage, collision, soldier attacks, and overkill beyond the victim's prior HP never count. Total eligible damage from that Main in the same tick is aggregated. At Main current HP <=50, healing is min(0.25 x eligibleDamage, 3); above 50, min(0.20 x eligibleDamage, 2). A shared 1.0-second cooldown prevents more than one grant in the interval. Healing is clamped to 100 HP.

When two Mains receive lethal damage during the same tick, both are marked defeated before any post-damage healing; the result is a draw. A Main cannot revive through healing on its lethal tick. Threshold comparison uses the surviving Main's HP **after** incoming damage but **before** permitted healing. This is deterministic and closes death/heal race conditions.

8. Soldier Generation and Reinforcement Economy

8.1 Wave scheduling and health boundaries

Both teams start with four soldiers. The first reinforcement wave occurs at simulation time t=20 s; subsequent waves occur at t=40, 60, 80 s, etc. At each wave the game samples current Main HP% = 100 x currentHP / 100. Rates use exhaustive intervals: **0 <= HP% < 10: 0**; **10 <= HP% < 40: 3**; **40 <= HP% < 80: 4**; **80 <= HP% <= 100: 5**. Dead Mains receive zero.

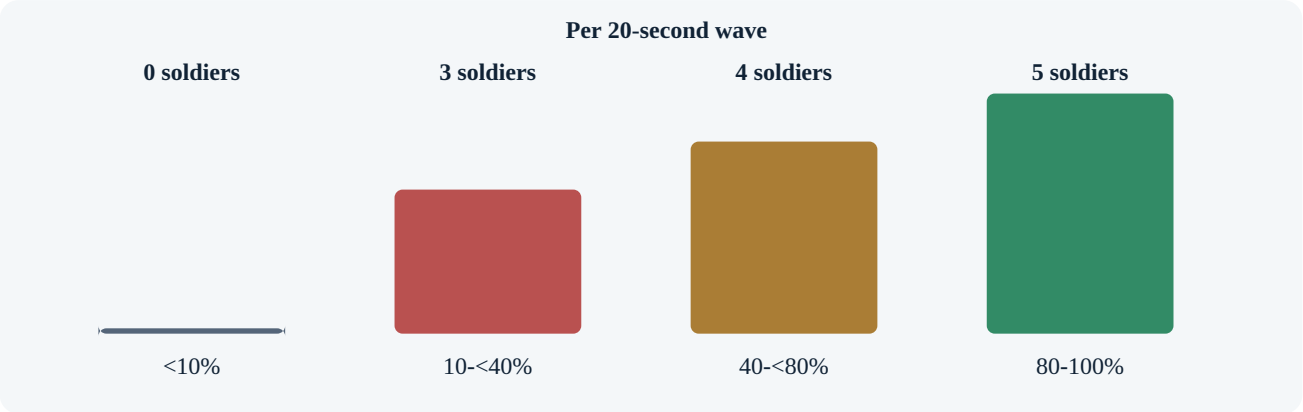


Figure 8. *Reinforcement quantity per 20-second wave according to exact Main-health bands.*

8.2 Population and capacity logic

Let N be the current number of living soldiers (0 to 50), B the base rate, U the Main's active streak bonus (0 or 1), and P the opposing active penalty (0 or 1). For B=0, adjusted reinforcement is zero. For B>0, adjusted = max(1,B+U-P), and actual spawn = min(adjusted,50-N). No backlog accrues when capacity or spawn geometry prevents a wave. The existing army is never reduced by a streak penalty.

8.3 Physical reinforcement locations

Each team owns a symmetric bounded reinforcement area placed away from direct opposing spawn. Wave units appear only at valid, unoccupied points with sufficient clearance from obstacles and actors. Deterministic placement order uses unit ID and a match-seeded candidate sequence. If the area is fully blocked, spawn only what can safely fit; the rest of that wave is discarded rather than accumulated. Players cannot force reinforcement on the Main's current battlefield position.

8.4 No-loss accumulation analysis

Main HP band	Soldiers/wave	Waves from 4 to reach >=50	Earliest theoretical time
<10%	0	Cannot accumulate	No finite time
10-<40%	3	16	320 seconds
40-<80%	4	12	240 seconds
80-100%	5	10	200 seconds

The values above assume zero casualties, a fixed health band, valid spawn slots and no streak modifiers. They are arithmetic bounds, not observed match durations. More aggressive high-health waves could still amplify early leads; mirrored battle data must quantify whether the 5:3 rate ratio is acceptable.

9. Kill-Streak Reward and Reinforcement Penalty System

9.1 Trigger and credit definition

A Main triggers a kill streak after **four directly credited enemy soldier eliminations within a rolling 30-second window**, provided it has no active effect and its 90-second activation cooldown has ended. Enemy Main eliminations terminate the match and are ineligible. Soldier kills and assists do not count. Kills by shield/body collision do not count; only that Main's rifle or sword direct damage can receive credit. Kills from the same attack may contribute multiple qualifying eliminations in deterministic actor-ID order.

For multiple lethal damage contributors within the same tick, kill credit goes to the actor that contributed the largest **applied** body damage on that tick. If tied, sort by stable damage-event sequence and then stable actor ID. Only a surviving Main may trigger a streak. Tracking is disabled while its effect or cooldown is active; when the cooldown finishes, the tracking queue starts empty, so kills cannot be banked during lockout.

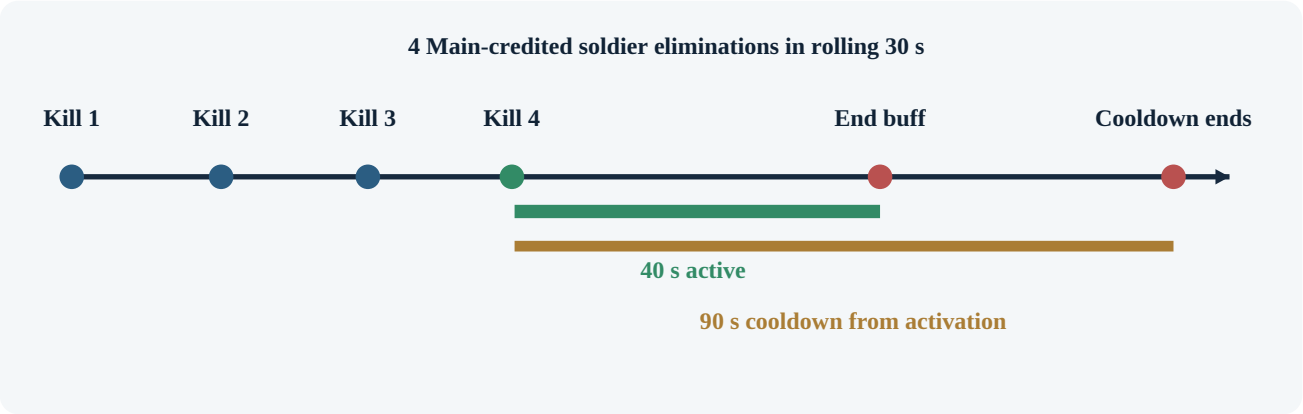


Figure 9. *Four Main kills within 30 seconds activate a 40-second reward and 90-second cooldown.*

9.2 Activation, duration, cooldown and expiration

The streak activates at the qualifying kill tick t0. Its active interval is **[t0, t0+40 s)**; cooldown is **[t0, t0+90 s)**. The shield reward is applied immediately at activation. Reinforcement modifiers apply only to waves whose event time falls inside the half-open effect interval. At t=t0+40, expiry and shield clamp execute **before** processing any reinforcement wave at that timestamp. At t=t0+90, a new four-kill tracking window may begin; the prior queue is empty.

9.3 Rewards and bilateral effects

Effect	Rewarded Main	Opposing Main
Reinforcements while active	+1 per eligible nonzero base wave	-1 per eligible nonzero base wave
Minimum if base >0	1 soldier per wave	1 soldier per wave
If base =0	Always 0	Always 0
Temporary max shield	+5 HP (25 -> 30)	No change
Immediate current shield	+5 HP (clamped to 30)	No change
Duration	40 seconds	40 seconds
Reactivation	90 seconds from activation	Own independent 90-second cooldown

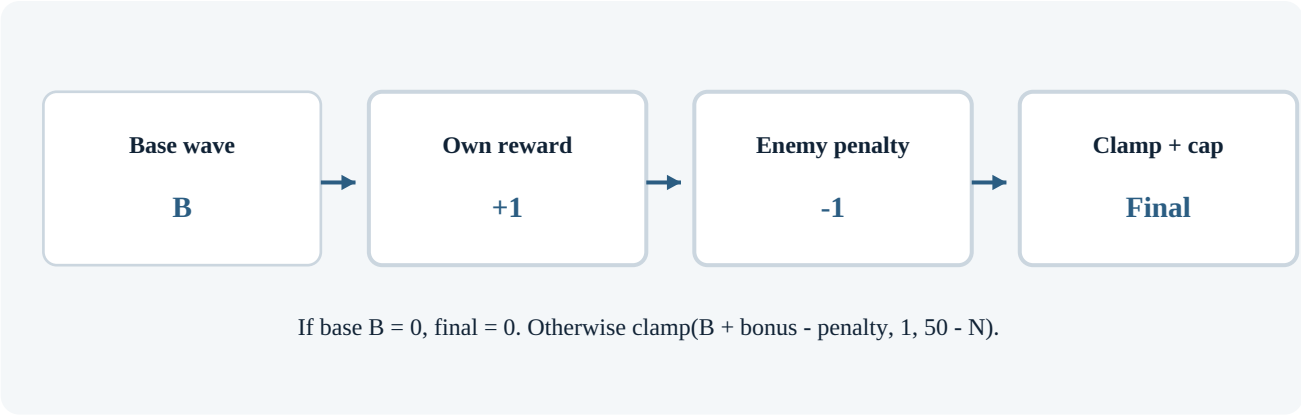


Figure 10. *Wave modifiers are additive then clamped; baseline zero and 50-living-soldier cap always override.*

9.4 Simultaneous streaks and modifier ordering

Each Main has its own independent tracker, reward interval and cooldown. Both may be active at once. For each team compute its own U=1 when rewarded and P=1 when the opponent's reward is active. When both are active, +1 and -1 cancel, so waves use the base health rate. Two active effects do not stack shield buffs: each Main may have only its own single +5. Existing soldiers are never removed, transferred, or converted.

9.5 Analytical reinforcement exposure

Base B	Reward only	Penalty only	Both active
0	0	0	0
3	4	2	3
4	5	3	4
5	6	4	5

A 40-second half-open effect on a global 20-second wave schedule normally intersects exactly two wave instants while the match is ongoing (unless activated near a match end). For uncapped teams with base B>0, the attacker gains at most two extra soldiers and the defender loses at most two expected spawns: a **four-soldier differential** over two waves. At a cap of 50 or with blocked spawn slots, impact is smaller. This counts wave **flow**, not an automatic four-unit instantaneous swing.

9.6 Snowball risk and gated alternatives

A rewarded healthy Main already gains reinforcement advantage, potential Main healing, and a +5 shield reserve. The system can amplify momentum. If mirrored seeded tests demonstrate unacceptable lead conversion (e.g. early streak side wins >70% across sufficiently many matched games), test in order: a 30-second effect; +1 reward without enemy -1 penalty; or a higher five-kill threshold. Do not enable those alternatives in the official v0.3 configuration unless the balancing process explicitly revises the constants.

10. Squad Creation, Management, and Explicit Commands

10.1 Dedicated squad ownership

The player may select one or multiple unassigned or assigned soldiers, create a numbered squad in slots 1-9, transfer members, remove members, choose a leader, set a rally, designate targets, and select formations and combat doctrine. Each squad holds at most ten soldiers. The complete army still caps at 50; unused squad capacity does not increase that cap. A soldier belongs to at most one numbered squad at a time. All others are unassigned reserve.

10.2 Orthogonal order components

Order field	Available values	Execution semantics
Movement	Move/Rally, Follow, Hold	Destination, leader or fixed guard area
Combat doctrine	Defend, Press, Harass, Disengage	Engagement rules; exactly one active
Formation	Compact, Line, Spread	Spacing/relative targets
Target preference	Unit, squad, area, none	Priority when legally visible
Fallback	Rally point or Main	Used when target/leader/path invalid

Movement, doctrine, formation, and target are independent slots; changing one leaves others untouched unless conflict resolution explicitly dictates otherwise. Disengage temporarily overrides offensive pursuit, moving the squad to its fallback while allowing obstacle avoidance. Commands persist across ordinary loss of sight, but automatic attacks stop until target visibility returns.

10.3 Doctrine constants

Doctrine	Damage	Forward speed	Shield damage taken	Engagement
Defend	1.00x	1.00x	1.00x	Counterattack within guard radius 8 m
Press	1.15x	0.85x	1.25x	Sustain attack on visible target
Harass	0.80x	1.25x	1.00x	Withdraw after 2 hits or 3 s
Disengage	No initiated attack	1.00x	1.00x	Return to fallback point

Damage multipliers apply after soldier base (0.20) and range falloff. Press shield multiplier applies to incoming shield-durability consumption as defined in Section 7, not additional body damage. Harass movement +25% is followed by the same 30% backward reduction as any soldier. All doctrine parameters are provisional but fixed for the alpha baseline.

10.4 Follow and leader failure rules

A Follow target must be living, allied, not self, and must not create a directed follow cycle. Group leaders must be current squad members. When the leader dies, select a replacement by lowest living unit ID; if no replacement exists, the squad becomes empty. On Follow target loss, switch movement to the saved rally point (or Main as default) without changing doctrine. Hold position references an area, not a nonexistent character.

10.5 Global Q/E stance

Q defensive stance allocates up to **five eligible unassigned soldiers** to bodyguard positions; E offensive stance allocates up to **two**. Other unassigned soldiers follow appropriate defensive or offensive autonomous behavior. Dedicated squad orders MUST never be silently canceled, and stance changes MUST NOT requisition assigned squad members. The policy changes immediately while physical movement occurs over time. Bodyguard reassignment has a 1.0-second lockout.

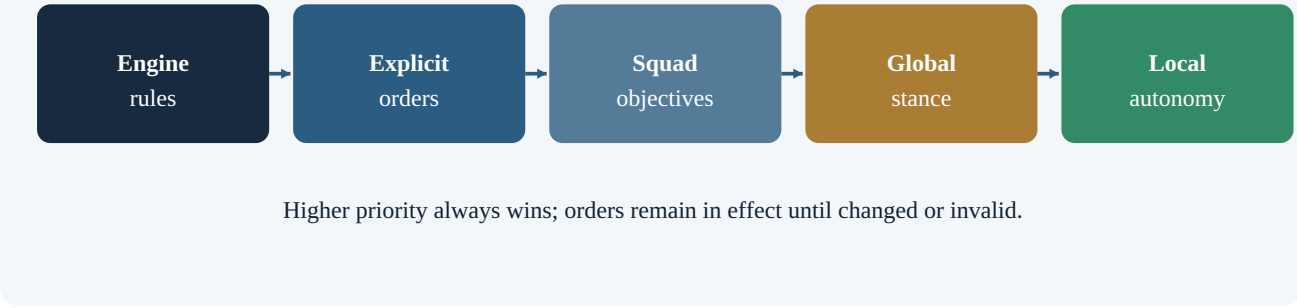


Figure 11. *Authority flows from mandatory rules to explicit orders and only then to autonomous choices.*

11. Soldier Autonomous Behavior and Formations

11.1 Behavior state model

The local controller tracks one primary task state: Idle, Moving, Following, Guarding, Seeking, Engaging, Harassing, Retreating, Regrouping, or Defeated. Equipment has independent states (Rifle Ready/Reloading, Sword Ready/Overloaded). This avoids combinatorial state explosion and enables a retreating soldier to reload without changing its movement obligation. Only legal transitions are accepted.

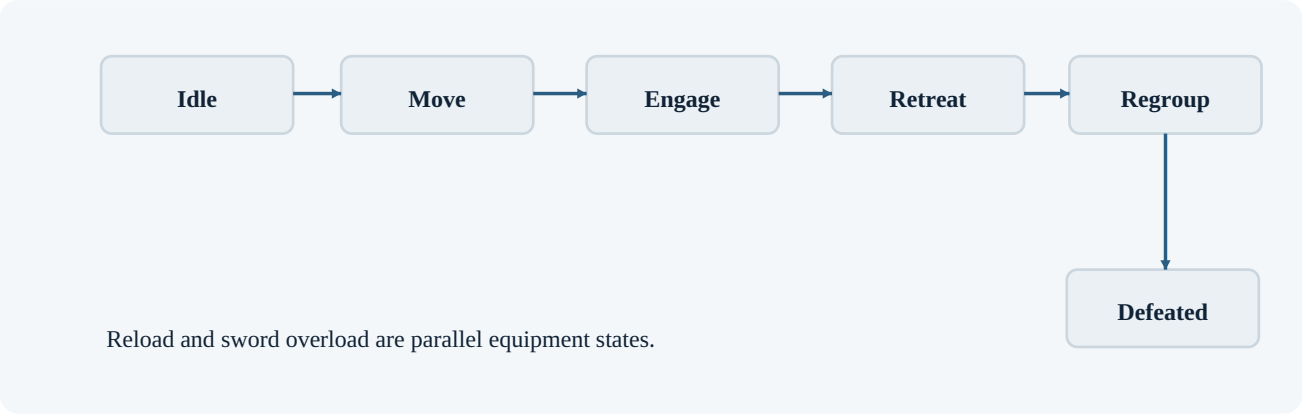


Figure 12. *Representative local unit behavior cycle; equipment substates run in parallel.*

11.2 Sensing and target selection

A soldier selects only currently visible enemies inside its 220-degree FOV and 24 m sight radius. Valid targets are ranked deterministically by active squad preference, threat, range, reachability and blocking risk. Selection hysteresis of 0.5 seconds reduces rapid target thrashing, except when an immediate threatening actor invalidates the current target. Last-known locations can guide searches but not enable direct lock-on attacks through cover.

11.3 Formations and no-stacking policy

Compact uses desired centers 1.0 m apart; **Line** 1.5 m; **Spread** 2.0 m. All include actual collider diameter and separation constraints. Formations rotate around a squad anchor and adapt to passage width. No formation permits intersecting collision bodies or firing through friendly actors. Squad members may temporarily break formation for safe obstacle avoidance and regroup after passage.

11.4 Combat routines

Soldiers align to a target using capped rotation speed; fire only with ammo, cleared cooldown, present visibility and an unobstructed predicted firing lane. Where a friendly body blocks the ray, soldier SHOULD hold fire and choose an offset rather than spend a bullet. If range is appropriate and strain permits, a soldier MAY switch to sword mode; its damage multiplier remains 0.20. A soldier with no ammo may retreat to resupply or engage in melee based on doctrine and danger.

11.5 Harass, retreat and regroup

Harass retires from a local engagement on its second successful hit or once 3.0 seconds of engagement have elapsed, whichever occurs first; engagement timer resets only after withdrawal to a minimum 6 m from the last opponent or full regroup at rally. Retreat is not teleportation. Disengage initiates no new attacks and routes to fallback. Defend can counter an actual threat within 8 m but cannot chase indefinitely beyond the guard radius.

11.6 Bounded path failure recovery

A local path request attempts available grid/mesh routes with a bounded 150 ms planning budget; when no path is returned, the soldier enters Hold/Regroup, reports "blocked", and retries after 0.5 s with exponentially bounded delays up to 4 s. A failed command remains visible in the squad HUD. No unit may execute illegal movement through cover while seeking a path. A command to an unreachable endpoint does not corrupt the last valid destination.

12. SLAI Artificial Intelligence Architecture

12.1 Separation of responsibility

The authoritative Game Engine owns physics, damage, collision, LoS, spawning, cooldowns and victory. Local soldier AI performs time-critical movement and targeting. Squad tactical controllers coordinate current objectives and formation updates. SLAI performs strategic evaluation, planning and adaptation through validated requests. This separation avoids using expensive multi-agent reasoning for every simulation frame.

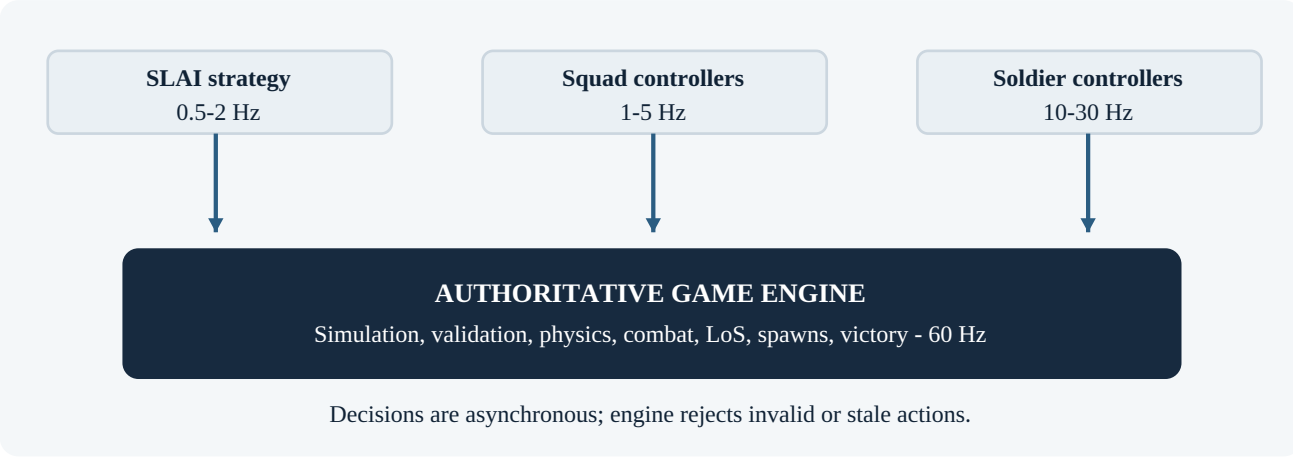


Figure 13. *Four layers preserve engine authority while allowing SLAI-guided tactical behavior.*

12.2 Agent participation and cost-aware dispatch

SLAI agent	Game responsibility
Perception	Summarize current authorized observations and confidence.
Planning / Reasoning	Select feasible strategic goals and compare tactical consequences.
Collaborative	Coordinate multiple independent squads and task delegation.
Learning / Adaptive	Offline tactic improvement; response to observed player patterns.
Knowledge	Retrieve validated tactics and previous match knowledge.
Evaluation / Quality	Outcome attribution, consistency checks, decision quality.
Safety	Check proposed action schema, legal constraints and runtime guards.
Observability	Decision timing, error rate, replay linkage and performance telemetry.
Orchestrator	Route only needed agents; bound latency and token/compute budgets.

12.3 Temporal update budgets

Controller	Design target	Required fallback
Authoritative game simulation	60 Hz fixed	Never waits on SLAI
Local soldier controllers	10-30 Hz	Maintain last legal policy
Squad controllers	1-5 Hz	Hold previous valid squad objective
SLAI strategy	0.5-2 Hz	Deterministic baseline strategy
Training/evaluation	Event-driven, post-match	No impact on active simulation

The frequencies are target update budgets, not established SLAI throughput. SLAI requests may execute asynchronously and MUST carry generation IDs and deadlines. On timeout or malformed output the engine continues the last valid order or switches to a deterministic baseline. A slow SLAI response cannot freeze the player, desynchronize simulation events, or directly modify game-state data.

12.4 State and action interface schema

The observation contains: match/time/version/seed identifiers; authorized own state; visible enemy entities; timestamped last-known entities; terrain and cover; squad roster/status; resources; reinforcement and streak timers; legal actions; computation budget. Every action contains: actor or squad ID, command type, legal target or destination, optional formation/doctrine, observation generation, issuedAt, expiresAt, and optional confidence. Missing identifiers and stale generations are rejected.

12.5 Strategic reasoning example

If SLAI Main falls to 35 HP against a player at 70 HP and is outnumbered 12-to-18, SLAI may assign eligible bodyguards through Q, choose Harass for a small squad, order a retreat to cover, and use local soldiers to screen. It may investigate a frozen last-known player position but cannot track the player behind opaque walls. Every output undergoes resource, visibility, ownership, and timing validation before application.

13. AI Decision-Making, Training, and Fairness

13.1 TacticalDecisionPolicy proposal

TacticalDecisionPolicy is a proposed locally trainable model, not a currently trained or deployed model. It ranks legal team/squad objectives such as Protect Main, Reposition, Press visible target, Harass flank, Contest Supply, Hold, Regroup and Disengage. The policy produces scores or preferences; a legal-action masker and deterministic resolver select executable commands. It does not predict or authoritatively set damage or positions.

13.2 Inputs and controlled outputs

Input group	Features / constraints
Own resources	Main HP/shield/ammo/strain; wave countdown; active streak windows.
Squad state	Counts, average HP, formation, doctrinal state, reachability, ammunition.
Enemy observations	Only visible entities and expired/active last-known records.
Map state	Cover geometry, supply location, legal routes and relative control.
Outputs	Legal tactic ID, squad ID(s), permitted target/rally, bounded duration, confidence.

13.3 Data collection and training regime

Start with deterministic scripted opponents and human recorded decisions, logging observation/action/outcome tuples under fixed schema and explicit match seed. Partition training, validation and holdout data by **match session** and where possible by map and policy version to avoid leakage. Train offline initially with supervised imitation and ranking; consider offline RL and self-play only after clean legal-action interfaces and reproducible baselines exist.

Potential reward components include win/loss; surviving Main HP; preserved soldier count; meaningful objective control; tactical damage dealt versus received; reduced invalid actions; and resource use. Weight values must be tuned to prevent reward hacking, such as indefinite shield camping or intentionally delaying victory. Never treat raw damage as a complete proxy for tactical intelligence.

13.4 Fairness and difficulty

SLAI and local soldier AI use identical collision, damage, weapon spread, vision, resource and reinforcement laws as the player. AI difficulty can change planning horizon, controlled reaction latency, targeting heuristics, and squad coordination accuracy. It MUST NOT see hidden current enemy positions or receive stealth buffs to weapon damage, spawning, or health. Mirrored self-play and independent baselines quantify unfair side and starting-position advantages.

13.5 Model release and fallback

Version each training dataset, observation/action schema, policy weights, evaluator thresholds and opponent cohort. Release only after holdout benchmarks, legal-action conformity checks and rollback capability. If model load, inference, validation or SLAI orchestration fails, use a deterministic scripted tactical policy with documented priorities. Match replay records emitted decisions rather than requiring inference reproduction.

14. Match Structure, Gameplay Progression, and Victory Conditions

14.1 Match initialization

Two Mains begin with 100 HP, 25 shield HP, zero sword strain, 10 loaded/40 reserve rifle rounds, and four newly created soldiers per faction. Each soldier begins with 30 HP, 8 shield HP and 6 loaded/18 reserve rounds. Default stance is Defensive (Q), and unassigned soldiers can serve as bodyguards. The common 20-second reinforcement and 30-second central supply schedules begin at simulation t=0. Both teams receive only authorized visible observations.

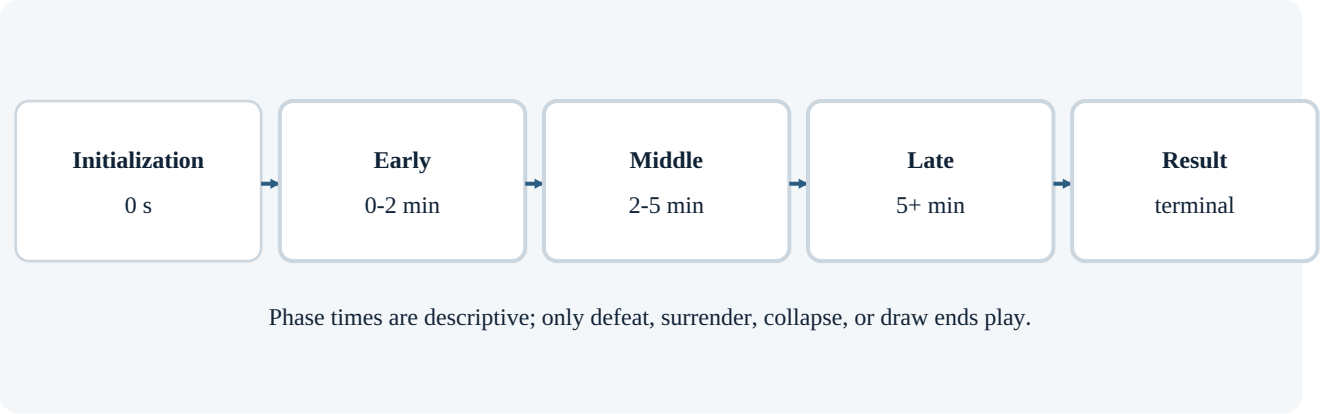


Figure 14. *Match phases are expected gameplay patterns, not hard transitions that change statistics.*

14.2 Progression by player pressure

Early play emphasizes Main dueling, reconnaissance and squad formation; middle play introduces more soldiers, flanking and contested supply; late play intensifies elimination opportunities, cover control, weakened Main protection and streak modifiers. No artificial hidden level-up multipliers are added as time passes. Soldiers and Mains keep the same base stats throughout a match, except explicit temporary effects.

14.3 Ammo supply and resupply

One central supply pickup becomes available at t=30, 60, 90 seconds etc. Only one may exist at a time. Main pickup restores up to 20 carried rounds, capped at 50; depleted pickups are consumed, and an uncollected pickup does not duplicate. Soldiers may refill six reserve rounds, not above their 24-round carried maximum, at their own reinforcement zone, once per soldier every 20 simulation seconds while alive. Ammo pickup cannot bypass loaded magazine size or the normal reload duration.

14.4 Terminal outcomes

The first valid defeat of an opposing Main ends the match. Explicit human surrender immediately concedes; SLAI automatic surrender remains disabled in this initial mode. Last Stand Collapse starts when Main HP is <10 **and** zero living allied soldiers, lasts 30 continuous simulation seconds, resets if either condition becomes false, and causes defeat if the condition persists at the end. A Main below 10% receives no reinforcement waves; a healing recovery to >=10% can restore future reinforcement eligibility.

14.5 Same-tick resolution and draws

Damage and death on the same simulation tick resolve before healing and before reinforcement. If both Mains die in that tick, declare a draw even if a wave or streak activation would have occurred at the timestamp. Terminal matches ignore subsequent queued squad and reinforcement events. If one Main is killed on the same tick as Last Stand expiry, the body-death result takes precedence for that Main; the winner is identical unless both simultaneously die, in which case draw takes precedence.

14.6 Pausing, disconnection and replay

Single-player pause freezes simulation time and all cooldowns, kill-window, shield regeneration and collapse timers. A temporary SLAI interruption retains baseline fallback and does not pause. Full save/restore of an in-progress match is optional beyond v0.3; match replay MUST preserve seeds, timestamped valid inputs and all authoritative events.

15. Controls, User Interface, and Player Feedback

15.1 Conflict-free keyboard/mouse map

Input	Context	Action
W / S / A / D	Always	Forward / backward / rotate left / rotate right
Left click	Normal combat	Primary rifle fire or Wide Swipe
Shift + left click	Sword equipped	Focused Thrust
Tab / R	Always	Switch mode / reload rifle
Q / E	Always	Defensive / Offensive global stance
1-9	Squads	Select numbered squad
Shift + 1-9	Selection exists	Create/replace squad membership assignment
Ctrl + left click / drag	Selection mode	Select unit / multiple soldier units
Right click terrain	Squad selected	Move/Rally to chosen point
Right click enemy	Squad selected, visible	Set target preference
F then right click ally	Squad selected	Follow selected ally; F enters target mode
Z / X / C / V	Squad selected	Defend / Press / Harass / Disengage
H / G	Squad selected	Hold / cycle Compact-Line-Spread formation
L then right click ally	Squad selected	Set leader from selected squad
Esc / Space	Always	Cancel targeting or menu / pause (single player)

Ctrl + click/drag is explicitly selection mode, so normal left click never accidentally selects instead of attacking. Shift+left click invokes sword thrust only while sword mode is active and does not interfere with Shift+number squad assignment. Right-click target orders require a selected squad and current visibility; hidden enemies cannot be designated by direct target-ID clicks. All inputs SHOULD eventually be remappable.

15.2 HUD minimum information

Top-level HUD must display Main HP/shield, active shield bonus, weapon/magazine/reserve, sword strain/overload, Q/E stance, living soldier count, reinforcement count/timer, streak four-kill progress, 30-second tracking window, active 40-second duration, and 90-second cooldown. Show Last Stand timer prominently when active. Squad cards show assigned number, roster, health condition, doctrine, formation, objective, target, engagement/retreat status, and "blocked" indicator.

15.3 Visibility and explanatory feedback

Render enemy sprites and exact HP only while currently visible to the allied team. An enemy lost behind cover should show at most a clearly marked stale location for three seconds. Threat indicators MUST NOT reveal exact hidden locations. At each wave, display the arithmetic: Base N, own streak +1, enemy streak -1, capacity/space effect, and actual soldiers received. This is essential for player trust.

15.4 Visual and accessibility conventions

Green/red faction identity should also use distinct icon outlines, selected-unit rings and squad glyphs so color vision deficiencies do not erase team identity. A shield block, body hit, friendly-lane block, out-of-range attack, sword overload, invalid squad target and streak activation need different legible feedback. Effects SHOULD remain readable at a 50-soldier density without obscuring the Main.

16. Gameplay Balance, Mathematical Analysis, and Exploit Prevention

16.1 Raw damage-per-second baselines

Ignoring spread, shields, reload, movement and obstruction, close-range rifle sustained fire between reloads yields **15/0.8 = 18.75 HP/s** for a Main and **3/1.5 = 2.00 HP/s** for one soldier. At 18 m: Main 2/0.8 = 2.50 HP/s; soldier 0.4/1.5 = 0.267 HP/s. Fifty soldiers theoretically firing point-blank every 1.5 s produce 100 HP/s before reload/accuracy/physical lane restrictions. This is an upper bound, not a realizable all-hit case in a constrained formation.

16.2 Analytical range/accuracy estimate

For a circular target radius r centered at distance d, a single target-centered unobstructed hitscan with uniform +/-a angular spread has approximate hit probability **P = min(1, atan(r/d)/a)**, with both angles in consistent units. This approximation assumes the entire target is within firing range, constant target position at the fire tick, no shields, and no obstacle/intermediate friendly collision. It is not a playtest result.

12 m aiming at...	Main 1.5 deg stationary	Main 2.5 deg moving	Soldier 4 deg stationary	Soldier 6 deg moving
Main (r=0.50 m)	100%	95.4%	59.7%	39.8%
Soldier (r=0.35 m)	100%	66.8%	41.8%	27.8%

Note the stationary Main reaches almost certain center-aim hits against both collider sizes at 12 m under this provisional model. This could make the Main unusually precise. Accuracy tests SHOULD compare 1.5-degree dispersion against slightly larger half-angle alternatives if combat proves too deterministic. No unseen accuracy nerf is included in v0.3.

16.3 Theoretical time-to-defeat examples

A fully accurate stationary Main needs seven unshielded point-blank rifle hits to defeat a 100 HP Main (ceil(100/15)=7); firing first at t=0 yields the seventh at t=4.8 s, excluding animation/collision/defense. Against 25 shield plus 100 HP, nine 15-HP hits can overcome 125 combined HP, with the ninth at t=6.4 s. If the Main reloads or moves behind cover, these times increase. A single soldier with only 24 rifle rounds cannot deal enough 3-HP body damage to defeat an unshielded 100-HP Main without resupply or sword assistance.

16.4 Healing and streak pressure

A point-blank Main rifle shot against full exposed body HP earns 3 HP healing when Main <=50% (25% x 15 = 3.75, capped at 3), or 2 HP when >50% (20% x 15 = 3, capped at 2). At most one healing event per second can occur, independent of multihits. A single full 40-second streak normally affects two waves and can at most add two allied soldiers and deny two opponent reinforcement soldiers before caps, creating a four-soldier relative swing.

16.5 Primary exploit register

Threat / root cause	Countermeasure	Verification
Spawn snowball / health scaling	Bounded 5:3 positive-wave ratio and strict cap	Mirrored first-streak win conversion
Streak farming / repeat triggers	Four direct kills, rolling 30 s, cooldown 90 s, no banking	Deterministic scripted kill timeline
Buff stacking / expiry race	One effect per Main, half-open 40 s, expiry before wave	Boundary ticks at t0+40 and t0+90
Healing death-reversal / overkill	Death before healing; applied-body-only capped heals	Simultaneous lethal shots
Wallhack / blind target tracking	Current LoS or stale 3 s, no hidden target IDs	Occlusion/reacquisition harness
Ally pass-through shots	Nearest collision harmlessly absorbs bullet	Allied line obstruction test

Threat / root cause	Countermeasure	Verification
Collision damage per frame	Distinct contacts + 1 s cooldown + separation	Sustained overlap 60 Hz
Formation stacking	Physical radii, separation, path and line restrictions	50-unit chokepoint stress
Rapid weapon-switch reset	Cooldown persistence and reload cancellation	Repeated Tab/R fire trace
Shield camping / infinite defense	Regen attack reset; objective/pickup pressure	No-attack 10-min adversarial matches
AI forbidden commands	Validate action schema, observation generation, authority	Fuzz invalid/stale actions

16.6 Balance decision gates

Balance changes MUST follow instrumentation: reproduce the defect; verify simulation logic; test visibility/collision and AI policies; then change the smallest numeric parameter necessary. Evaluate confidence intervals rather than treating a small set of matches as proof. Candidate targets: typical matches 6-10 min; no systematic spawn-side win bias outside a 45-55% pilot band after sufficient trials; zero accepted illegal AI actions; and meaningful positive incremental value of both Main and squad skill.

17. Technical Requirements and Implementation Architecture

17.1 Fixed-step authoritative simulation

The engine runs 60 simulation ticks per second with simulation-tick timestamps. Pauses suspend simulation time. Events that share a tick use deterministic processing order. Visual rendering and SLAI inference can run at independent frame/frequency rates, but neither may advance authoritative physics or apply damage outside the fixed tick.

17.2 Tick order and event ownership

At each tick: (1) expire buffs/cooldowns whose time elapsed; (2) validate commands and sample movement inputs; (3) update movement/rotation and collision; (4) refresh visibility when perception budget permits, and validate attacks against present geometry; (5) resolve damage and shield interception; (6) mark deaths and determine attribution; (7) apply eligible healing to surviving Mains; (8) activate streaks from valid credited soldier kills; (9) process wave and supply events; (10) evaluate Last Stand and terminal conditions; (11) emit observation/replay telemetry. The terminal double-Main-death check after damage always overrides later wave benefits.



Figure 15. *High-level pipeline; steps within a tick use stable ordering and tie breaks.*

17.3 Data types and invariant enforcement

Entity / record	Minimum required fields
Actor	stable ID, team ID, position, orientation, HP, shield, alive flag
Weapon	mode, ammo, reserves, fire/reload timers, strain/overload
Squad	roster IDs, leader, movement, doctrine, formation, target, rally
VisibilityRecord	observer/team, target, observedAt, last-known position, visible flag
StreakState	credited kill timestamps, activatedAt, expiresAt, cooldownEnd
MatchState	seed, fixed tick, schedule, actor lists, winner/result reason
AICommand	source generation, actor/squad, command, target, deadline, status
ReplayEvent	tick, deterministic ordering key, event data, schema version

17.4 Performance and reproducibility

Target 100 living soldiers plus 2 Mains, projectiles represented as raycasts, and collision/LoS interactions on the selected target platform. Use spatial broad-phase grids, limited-neighbor detection, cached waypoints, capped LoS sample rays, and bounded path replanning. SLAI decisions are asynchronous and should be batched at squad level. A fixed seed, stable iteration order and logged input/tick sequence should enable deterministic replays when the chosen engine supports deterministic arithmetic.

18. Testing, Validation, and Development Roadmap

18.1 Mechanical test matrix

Test group	Scenarios	Acceptance checks
Direct duel	Full/empty shield, 2m/18m, still/moving	Damage within formula; cooldowns exact
Unit combat	1v1, 1v5, 10v10, 50v50	No overlaps, valid target/hit behavior
Cover and LoS	Fully/partly hidden, corner, moving cover approach	No through-wall visibility/shots
Accuracy	Stationary/moving, seeded repeat, varied radius	Same seed = same ray results
Reinforcement	All HP boundaries, capacity 49/50, blocked area	Wave count exactly clamped
Streak	Four kills/30s, expiry, both active, cooldown	Reward 40s; recharge lockout 90s
Collision	Overlap, shield-face, rapid recontact	Single event per eligible contact
Commands	Leader death, cyclic Follow, invalid target	Deterministic fallback without override
SLAI failure	Timeout, stale ID, lost model	No freeze; baseline opponent works
Terminal logic	Double KO, no soldiers, heal across 10%	Correct result/timer reset

18.2 Quantitative experiments

Run symmetric map tests with scripted bots, alternate faction spawn sides, and report by weapon, stance, squad doctrine, health band and streak occurrence. Collect Main DPM, soldier DPM, hit fraction by distance, LoS rejection counts, shield absorption, healing HP, army counts per 20 seconds, squad survivability, match length, and win rate. Use at least hundreds of seeded games for early directional balance evaluation, and confidence intervals to determine whether more runs are needed.

18.3 Staged roadmap

Milestone	Deliverables	Exit test
A: Main duel	Movement, shield, rifle, sword, collision, defeat	100 deterministic direct duels reproducible
B: Unit battle	Soldier AI, reinforcements, formation baseline	Stable 50v50 with no population breaches
C: Squad UX	Nine squads, orders, formation and HUD	All explicit command edge cases pass
D: Information	LoS, hidden sprites, last-known, cover	Geometric visibility test suite passes
E: Streak system	Kill credit, timers, buffs, penalties	Simultaneous / edge case fixtures pass
F: SLAI	Structured interface, policy routing, fallback	Legal output and timeout stability
G: Balance	Analytics, mirrored seeded matches, human testing	Documented metric-driven tuning decisions

18.4 Definition of done

The core alpha is acceptable when two teams can finish complete matches; no health/ammo/squad cap is violated; all damage and LoS rules are deterministic under seed; wave arithmetic, kill credit and streak effect boundaries are consistent; every squad doctrine has verifiable behavior; SLAI obeys authoritative input/output contracts; invalid AI responses fail safely; telemetry supports seed replay; and early human testing confirms both action and tactical control are usable.

19. Outstanding Design Risks and Recommendations

19.1 Prioritized unresolved risks

Priority	Risk	Evidence needed / next action
P1	Main stationary rifle near-perfect center-aim hits at common ranges	Compare target sizes/angles using recorded duels
P1	Shield direction unfamiliar to players	Usability test retreat orientation with novices
P1	Reinforcement and streak snowball	Mirrored seeded tests by first-streak timing
P1	SLAI latency and invalid action rates	Profile decision deadlines with 100 soldiers
P2	50-unit collision and chokepoints	Worst-case map/pathfinding stress
P2	UI command overload	Task-time and error-rate user tests
P2	Supply pickup encourages overcentralized fighting	Compare no-supply and single-supply arenas
P2	Soldier sword damage too low for meaningful melee	Measure exhausted-ammo combat end states
P3	Future persistent progression balance	Keep cosmetics non-stat-affecting

19.2 Decision log and change trace

Parameter	Original / v0.2	v0.3	Reason / validation
Rear shield	Rear; 120 deg in v0.2	120 deg rigid rear arc	Preserve identity; geometric collision tests
Backward speed	Direction only; v0.2 0.75x	Exactly 0.70x	Explicit mandate; speed assertion tests
Main / soldier accuracy	Undefined	+/-1.5/2.5 deg vs +/-4/6 deg	Fair precision distinction; duels
Visibility	Undefined	24m/220deg, ray LoS, 3s stale	No information cheating; occlusion tests
Waves	Original 0-6; v0.2 0/3/4/5	0/3/4/5 unchanged	Reduced win-more scaling; tournament test
Streak	Not defined	4/30s; +1/-1 wave; +5 shield; 40s; 90s CD	Rewards Main skill; snowball test
Dedicated squads	Original Shift+1-9; v0.2 nine groups	Nine <=10, stable orders/roles	Hybrid action-strategy parity; UX tests
Sword	Original permanent break; v0.2 overload	6s overload then 40 strain	Avoid irrecoverable combat failure; test
Healing	Original 25%/20%; v0.2 caps 3/2	25%/20%, 3/2 caps and 1s CD	Preserve recovery while bounding exploit
Font	Requested Times New Roman	Tinos (disclosed fallback)	Exact font unavailable in environment

19.3 Change governance

The numeric values in this document are the single alpha baseline. Altering them requires a versioned change record, a stated design hypothesis, a relevant regression test and comparison against prior results. For critical mechanics, update both the main body and Official Ruleset together. The prioritized next engineering effort is a combat/LoS prototype, not a full-trained SLAI rollout.

R. Official GangUP Ruleset - v0.3

This section defines the **normative alpha rules** independently of explanatory game-design discussion. "MUST" and "MUST NOT" are mandatory; "MAY" is optional. All specified numbers are authoritative v0.3 prototype constants, though subject to future formally versioned balance changes.

R1. Teams, purpose, entities

R1.1 A match MUST contain exactly two opposed factions, one living Main per faction at initialization, and four living soldiers per faction. The number of living soldiers per team MUST NOT exceed 50. Soldier/Minion refer to the same entity class.

R1.2 A team MUST win when its opponent Main is defeated, legally surrenders, or suffers Last Stand Collapse, except that simultaneous Main defeats in one simulation tick MUST produce a draw. AI automatic surrender MUST remain disabled in the base game.

R2. Attributes and movement

R2.1 Main: 100 body HP, 25 shield HP, base forward speed 4 m/s, collision radius 0.50 m. Soldier: 30 body HP, 8 shield HP, base forward speed 6 m/s, collision radius 0.35 m. Soldier attack multiplier MUST be 0.20 before doctrine modifiers.

R2.2 W/S MUST move Main forward/backward in facing space, A/D MUST rotate it; weapon is forward and shield is exactly rear-facing. Main base angular speed MUST be 180 degrees/s, soldier maximum 240 degrees/s. Provisional acceleration limits are 12 m/s² Main and 16 m/s² soldier.

R2.3 Modified forward speed MUST equal base speed multiplied by current doctrine speed factor; backward speed magnitude MUST equal precisely 70% of that modified forward speed. Press factor=0.85; Harass=1.25; other doctrines=1.00. Movement MUST be blocked by solid obstacles and collision bodies.

R3. Weapons, range and rifle accuracy

R3.1 Rifle Main damage MUST be 15 for d<=2 m; 15 - 13(d-2)/16 for 218 m. Soldier rifle MUST apply 0.20 multiplier before doctrine effects. Main fires at intervals >=0.8 s, soldier >=1.5 s.

R3.2 Main magazine MUST hold at most 10 and total carried bullets <=50. Soldier magazine MUST hold at most 6 and total carried <=24. A normal reload MUST last 1.5 uninterrupted simulation seconds and transfer reserve to magazine at completion. Mode switching and sword attack MUST cancel ongoing reload without reserve consumption; moving MUST NOT.

R3.3 Every legal shot MUST consume one magazine round and MUST use a seeded uniform angular dispersion sampled within +/-1.5 degrees (stationary Main), +/-2.5 (moving Main), +/-4.0 (stationary soldier) or +/-6.0 (moving soldier). Translational speed above 0.2 m/s means "moving". SLAI MUST NOT receive perfect accuracy.

R3.4 A shot MUST raycast from the muzzle up to 18 m and resolve only the closest eligible collision. Opaque solid obstacles block; allied bodies absorb shots harmlessly; shield interception precedes exposed body damage when correctly oriented. Friendly-fire body damage MUST remain disabled. Missed shots and ally-blocked shots still consume ammunition.

R4. Line of sight and hidden information

R4.1 All actors MUST obey 24 m vision range and a 220-degree forward-centered FOV. A target is currently visible when at least one of its nine designated collider sample points (center plus eight inset perimeter points) has a valid unobstructed LoS ray within range/FOV. Full obstruction MUST make the target invisible.

R4.2 A visible target MAY still be physically unhittable when muzzle geometry, spread, cover, shield or friendly body prevents damage. Opaque solid cover MUST block both LoS and projectiles. Automatic target selection MUST NOT use concealed target IDs.

R4.3 A lost enemy MAY retain a frozen last-known position for 3.0 simulation seconds. This record MUST NOT update while the enemy remains hidden and MUST NOT enable direct automatic attacks. Expired sightings MUST revert to unknown. Player display and SLAI observations MUST use equivalent team-valid visibility information.

R5. Sword and strain

R5.1 Wide Swipe MUST deal Main 7 HP per target, reach 2 m in a frontal 120-degree sector, affect no more than three targets (nearest first), and add 18 strain once after any valid hit. Thrust MUST deal Main 12 HP, reach 2.5 m in a narrow 20-degree sector, hit only one nearest legal target and add 24 strain after a hit. Soldier damage is 20% of these amounts. Both share a 0.85 s sword cooldown.

R5.2 A shield interception counts as a successful sword hit; a complete miss grants zero strain. Strain MUST be within 0-100. It recovers at 12 points/s beginning 1 s after the latest sword attack attempt. At 100 the sword MUST overload for 6 s, disable sword attacks and then reset strain to 40. Rifle usage MAY continue. Neither weapon mode switch MAY reset attack cooldowns.

R6. Shield, contact and healing

R6.1 A live shield MUST cover a 120-degree rear arc, mathematically dot(defenderForward,incomingSourceDirection)<=-0.5. Shield HP absorbs covered damage before body damage. Shields MUST NOT independently rotate.

R6.2 Shield regeneration MUST start only after 8 continuous seconds without attack attempts, incoming damaging contact or damage received. Main recharge MUST be 2.5 HP/s, soldier 1.0 HP/s, capped at current shield max. Attacking even if missed MUST reset recharge delay.

R6.3 First eligible enemy unshielded body/body contact MUST damage both for 5 HP. Shield/shield contact MUST cost 1 shield HP to each. Shield/body contact MUST cost 1 to intercepting shield and 5 body HP to the exposed opponent. Re-damage requires both a 1 s pair cooldown and >=0.1 m separation/re-contact. Friendly and wall collisions MUST NOT deal impact HP damage.

R6.4 Only a surviving Main MAY heal from its own rifle/sword enemy body HP actually removed, never shields, collisions, assists, allied hits or overkill. Aggregate that Main's tick-eligible damage, grant at most once per 1 s: min(25% damage,3) if current HP<=50, else min(20% damage,2); clamp at 100. Death determination MUST precede healing.

R7. Reinforcements and resource supply

R7.1 Both teams MUST receive a scheduled reinforcement opportunity at t=20,40,60... seconds of simulation time. For Main HP% in [0,10): 0 soldiers; [10,40): 3; [40,80): 4; [80,100]: 5. Dead Main -> 0.

R7.2 A living soldier total N MUST remain <=50. If base B=0, adjusted wave MUST equal zero even with streak buffs. Otherwise final intended wave MUST equal max(1,B+ownBonus-opponentPenalty), then actual spawned MUST be capped by 50-N and valid unoccupied spawn positions. Unused slots from blocked wave MUST NOT accumulate.

R7.3 Reinforcements MUST appear in faction-owned valid reinforcement zones. A central pickup MUST appear at t=30,60,90... only if no earlier pickup remains; a Main collecting it restores up to 20 carried rifle rounds within its cap. Soldiers MUST be able to restore six reserve rounds at their reinforcement zone once per 20 s per soldier, not beyond maximum ammo.

R8. Main kill-streak effect

R8.1 A surviving Main MUST activate a streak after four direct enemy soldier kills by that Main's rifle/sword inside a rolling 30-second window when eligible. Allied soldier kills, assists, contact kills and Main defeats MUST NOT qualify. Simultaneous credited kills MUST be ordered by larger actual applied lethal-tick damage, then damage-event order, then actor ID.

R8.2 Activation at t0 MUST grant active effect during [t0,t0+40 s) and lock reactivation during [t0,t0+90 s). Tracking MUST remain disabled during active effect/cooldown and start with an empty queue at cooldown completion. Kills MUST NOT extend effects or bank toward later activation.

R8.3 While active the rewarded Main MUST gain +1 to eligible base-positive reinforcement waves, and the opposing Main MUST suffer -1 (min 1 when base-positive). Both simultaneously active effects MUST cancel wave differences. Rewarded Main maximum shield MUST temporarily increase from 25 to 30 and its current shield gain up to 5, clamped to 30. At expiry max MUST return to 25 and current MUST clamp to 25. Effects MUST NOT stack. At exact expiry time, remove buff before processing waves.

R9. Dedicated squads and global stances

R9.1 Each faction MUST offer nine independent assignable squad slots, each holding 0-10 soldiers. Soldiers MUST belong to at most one squad; remaining soldiers are unassigned reserve. A squad MAY specify leader, rally, movement, doctrine, formation and visible target preference. A transfer MUST update old and new rosters atomically.

R9.2 Supported movement commands MUST include Move/Rally, Follow and Hold. Combat doctrines MUST include Defend, Press, Harass and Disengage. Formations MUST include Compact (1.0 m spacing), Line (1.5 m) and Spread (2.0 m), subject to collision clearance.

R9.3 Press MUST modify damage by 1.15x, movement speed by 0.85x and incoming shield durability consumption by 1.25x without creating extra raw body damage. Harass MUST modify damage by 0.80x and movement speed by 1.25x and withdraw after two successful hits or 3 s engagement (whichever first). Defend MUST counterattack within an 8 m guard radius; Disengage MUST not initiate attacks and MUST retreat to fallback.

R9.4 Q MUST reserve up to five eligible unassigned bodyguards, E up to two. Stance reassignment lockout MUST last 1 s. Dedicated squad orders MUST NOT be canceled or their soldiers taken by global stance. Command hierarchy MUST be: mandatory engine constraints > explicit orders > current squad objectives > global stance > local autonomy. Cyclical Follow, self-follow and defeated target references MUST be rejected or safely resolved.

R10. Soldier behavior and SLAI authority

R10.1 Soldiers MUST perform collision avoidance, local pathfinding, aim alignment, legal target selection, reload, attack cooldown management, formation maintenance, regroup and retreat while retaining explicit player orders. No unit MAY cross blocking geometry or exceed legal modified speed.

R10.2 SLAI MAY choose Main/squad strategic goals only through validated actor/squad commands, target/destination, issued timestamp, expiry, observation generation and legal parameters. It MUST NOT modify physics, health, ammo, shield, wave rates, victory states, or target hidden enemies using forbidden exact data.

R10.3 The simulation MUST continue at fixed 60 Hz independent of SLAI. Local soldiers target 10-30 Hz, squad control 1-5 Hz, strategic SLAI 0.5-2 Hz. On AI timeout, stale generation, invalid action or model outage, the engine MUST reject the action and continue prior legal orders or a deterministic scripted fallback.

R11. Match timing, victory and same-tick cases

R11.1 Last Stand Collapse MUST begin only while Main HP<10% and allied living soldier count=0. Its continuous countdown MUST be 30 s and MUST reset whenever either condition becomes false. At expiration with both still true the team loses. A Main can recover beyond the threshold via eligible healing, and future waves then follow its current HP band.

R11.2 A Main reaching 0 HP MUST be defeated immediately when current-tick damage resolves. If both Mains die on the same tick, the result MUST be a draw and posthumous healing MUST NOT revive either. Terminal state MUST cancel future spawning/commands. A player MAY surrender through an explicit confirmation; no automatic SLAI surrender is enabled.

R11.3 All cooldowns, streak windows, shield recharge, supply and reinforcement timers MUST use simulation time, not wall-clock time. Single-player Pause MUST freeze these timers. Replay MUST record input decisions and authoritative event tick order with the seed.

R12. Non-normative future options

R12.1 The following are outside v0.3 mandatory rules: permanent sword breakage, additional units or factions, neutral recruitment, destructible/transparent cover, moving walls, multiplayer, stat-bearing persistent progression, and automatic AI surrender. These MAY be proposed later only as separately versioned features with balance and regression tests.